

NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Fifth Semester B.E. (ISE) (Credit System) Degree Examinations
 January 2023

20ISE34 – OBJECT ORIENTED MODELING AND DESIGN

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1. a)	What is object-orientation? Explain the different stages of OO methodology.	6	L*2	1	1
b)	Define the following terms with an example: i) Objects ii) Classes iii) Class Diagrams iv) Values and Attributes v) Operations and Methods	10	L1	1	1
c)	What is event? Explain single event and time event with example.	4	L2	1	1
2. a)	What is an association end? What are the properties of association end?	6	L1	2	1
b)	Construct the generalization for Equipment class and also write a note on generalization.	10	L3	2	1
c)	Draw the state diagram for telephone line with activities.	4	L3	2	1
3. a)	Outline constraints. Explain the constraints on objects, generalization sets and on links along with the uses of constraints.	10	L2	2	1
b)	Explain the state diagram behaviors with suitable examples.	10	L2	2	1
Unit – II					
4. a)	Describe the guidelines for the following: i) Use case models ii) Sequence models	10	L2	3	1
b)	What is concurrency? With suitable examples, explain the following: (i) Aggregation Concurrency ii) Concurrency within an Object (iii) Synchronization of concurrent Activities	10	L2	3	1
5. a)	Define nested state with suitable examples. Analyze the problems with flat state diagrams. Draw nested states for a telephone line system.	10	L4	3	1
b)	Explain the steps performed in construction in domain state model with an example.	10	L2	3	1
6. a)	What is a software development process? List and explain different stages in software development process.	10	L2	3	1
b)	What is a problem statement? Briefly discuss on the kinds of requirements.	10	L2	3	1
Unit – III					
7. a)	Classify the steps involved in constructing an application state model. Draw the state diagram for session controller.	10	L4	4	1
b)	Mention the steps involved in implementation modeling. Explain the first step in detail.	10	L2	4	1
8. a)	What are the steps to improve organization of class design? Explain.	10	L2	5	1
b)	Summarize the realization of association in a class diagram with suitable examples.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
